11. Sequences and Series

11.1. Sequences

11.1b Learning Objectives

- Understand how to write a sequence and find their formula if they have one
- Find the limit of a sequence using limit laws and theorems
- Determine whether sequences converge or diverge using theorems
- Determine whether a sequence is increasing or decreasing, and whether it is bounded

Theorem 11.1.1 Squeeze Theorem for Sequences

If $a_n \leq b_n \leq c_n$ for all $n \geq N$ where N is a positive integer, and

$$\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n,$$

then we must have

$$\lim_{n\to\infty} b_n = L$$

Example 11.1.1

Does the sequence $\left(\dfrac{\cos(n)}{n}\right)_{n=1}^{\infty}$ converge?

Solution

First, notice $-1 \leq \cos(n) \leq 1$. So,

$$\frac{-1}{n} \leq \frac{\cos(n)}{n} \leq \frac{1}{n}.$$

Since both $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ and $\left(\frac{-1}{n}\right)_{n=1}^{\infty}$ both converge to 0, then by the squeeze theorem, the sequence $\left(\frac{\cos(n)}{n}\right)_{n=1}^{\infty}$ converges to 0 as well.

Theorem 11.1.2 Limits of Continuous Functions

If $\lim_{n\to\infty} a_n = L$ and a function, f, is continuous at L, then

$$\lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right) = f(L).$$

Example 11.1.2

Does the sequence $\sin\left(\dfrac{\pi}{2} + \dfrac{1}{n}\right)_{n=1}^{\infty}$ converge?

Solution

Because the function $f(x) = \sin(x)$ is a continuous function, then by the theorem:

$$\begin{aligned}\lim_{n\to\infty}\sin\left(\pi+\frac{1}{n}\right) &= \sin\left(\lim_{n\to\infty}\frac{\pi}{2}+\frac{1}{n}\right)\\ &= \sin\left(\frac{\pi}{2}\right)\\ &= 1\end{aligned}$$

So the sequence $\sin\left(\frac{\pi}{2}+\frac{1}{n}\right)_{n=1}^{\infty}$ converges to 1.

Theorem 11.1.3

If $\lim_{n\to\infty}|a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Example 11.1.3

Does $\left(\frac{(-1)^n}{n}\right)_{n=1}^{\infty}$ converge?

Solution

Observe that

$$\lim_{n\to\infty}\left|\frac{(-1)^n}{n}\right| = \lim_{n\to\infty}\frac{1}{n} = 0,$$

so $\lim_{n\to\infty}\frac{(-1)^n}{n} = 0$ and the sequence $\left(\frac{(-1)^n}{n}\right)_{n=1}^{\infty}$ converges to 0 as well.

Definition 11.1.1

A sequence $(a_n)_{n=1}^{\infty}$ is said to be **increasing** if $a_{n+1} \geq a_n$ for each n.

A sequence $(a_n)_{n=1}^{\infty}$ is said to be **decreasing** if $a_{n+1} \leq a_n$ for each n.

A sequence that is either always increasing or always decreasing is called **monotonic**.

Definition 11.1.2

We say that the sequence $(a_n)_{n=1}^{\infty}$ is **bounded above** if there exists a number M such that $a_n \leq M$ for all n.

Similarly, we note that the sequence $(a_n)_{n=1}^{\infty}$ is **bounded below** if there exists a number m such that $a_n \geq m$ for all n.

We say that a sequence $(a_n)_{n=1}^{\infty}$ is **bounded** if there exists a number M such that $|a_n| < M$ for all n. That is, $(a_n)_{n=1}^{\infty}$ is bounded above *and* below.

Example 11.1.4

Determine if the sequence is increasing, decreasing, or neither. Determine whether the sequence is bounded:

$$(n)_{n=1}^{\infty}$$

Solution

$$(n)_{n=1}^{\infty} = (1, 2, 3, 4, 5, 6, 7, 8, 9, ...)$$

This sequence is(monotonically) increasing, and bounded below by 0. However, it is not bounded (because it is not bounded above).

Example 11.1.5

Determine if the sequence is increasing, decreasing, or neither. Determine whether the sequence is bounded:

$$\left(\frac{1}{2n+1}\right)_{n=1}^{\infty}$$

Solution

$$\left(\frac{1}{2n+1}\right)_{n=1}^{\infty} = \left(\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, ...\right)$$

This sequence is(monotonically) decreasing. It is bounded above by $M = \frac{1}{3}$ and it is bounded below by $m = 0$. Thus, this sequence is bounded.

Example 11.1.6

Determine if the sequence is increasing, decreasing, or neither. Determine whether the sequence is bounded:

$$\left(\frac{1-n}{2+n}\right)_{n=1}^{\infty}$$

Solution

$$\left(\frac{1-n}{2+n}\right)_{n=1}^{\infty} = \left(0, -\frac{1}{4}, -\frac{2}{5}, -\frac{3}{6}, -\frac{4}{7}, ...\right)$$

These fractions are a bit harder to tell what's going on so we will also use other tools to help us analyze this one. First, notice that every value after the first is negative. So, the sequence is bounded above by $M = 0$. Also notice the function $y = \frac{1-x}{2+x}$ has a horizontal asymptote at $y = -1$. We can see the fractions are indeed approaching $y = -1$ and that y is decreasing on $[1, \infty)$ (so we know it doesn't pass the asymptote and come back up towards it), so the lower bound is $m = -1$. The sequence is decreasing (monotonically) and is bounded.

Example 11.1.7

Determine if the sequence is increasing, decreasing, or neither. Determine whether the sequence is bounded:

$$((-1)^n)_{n=1}^{\infty}$$

Solution

This sequence is bounded above by $M = 1$ and below by $m = -1$, so it is bounded. It is not monotonic, it alternates between 1 and -1.

Theorem 11.1.4

Every convergent sequence is bounded.

Warning

These statements are not 2 way! This theorem says if we start with a convergent sequence, then we can conclude that it is also bounded. Alternatively, if we start with a bounded sequence there is NO guarantee that the sequence is convergent (see Example 11.1.7)

Note

We can add a condition to the above theorem to make it go "both directions"...

Theorem 11.1.5

If a sequence $(a_n)_{n=1}^{\infty}$ is both bounded AND monotonic, then the sequence converges.

Corollary 11.1.1

If a sequence is **unbounded**, then it diverges.

11.1b Section Summary:

- We looked into theorems and more definitions around sequences and their convergence.
- We can identify when sequences are monotonically increasing or decreasing and if they are bounded and use this information to (potentially) say more about if they converge or not.